Diagnostics

One screen per input or output. Focused content is large. Every detail page names the GPIO and Board A terminal and shows the live level. Supporting values stay small. Header prefix is `DIAG ·`.

Draft. Pixel layout: UI/diagnostics.html. Terminal numbers match Board A.

Both jog pages and the obstruction test own the motor. They carry an amber banner while live. Hall and rocker cannot preempt them. HOLD cancels: PWM off, current decay, K1 unchanged. A short PRESS while moving also drops PWM (sampled in the main loop, not an interrupt). **Jog by time** queues milliseconds per EC11 detent. While that move is running it does not read the shaft encoder, so a NACK on I²C cannot stretch the queue. Leftover queue time expires on its own. **Jog by position** queues relative shaft counts. It does not need a known origin. It runs only while the AS5600 ACKs and the magnet is in range; a lost encoder stops PWM. Opening that page polls the encoder.

1. Menu

Thirteen items. TURN windows the list. Pips flag anything already unhealthy. PRESS opens. HOLD exits to Main. Figure 1 is an older 12-item shot; live firmware splits jog into time and position. Items after the first window have their own figures below.

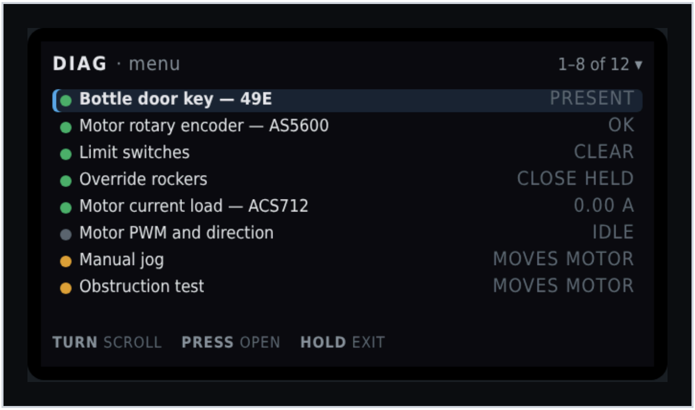

Figure 1. DIAG · menu. First window, bottle key selected. Live list also has Jog by time and Jog by position.

On screen	Meaning
`DIAG · menu`	Header left.
`1–6 of 13 ▾` (live) or the older `1–8 of 12` shot	Header right. Window into the 13-item list. The triangle means more items below.
Green pip + highlight + `Bottle door key — 49E` + `PRESENT`	Selected. Hall sensor. Value is live cooked state.
Green pip + `Motor rotary encoder — AS5600` + `OK`	I^2C ACK and shaft magnet in range. Amber pip + `no mag` means the bus is live but STATUS is lost, too weak, or too strong. Red pip + `no i2c` means the chip did not ACK at 0x36.
Green pip + `Limit switches` + `CLEAR`	Neither marker is made.
Green pip + `Override rockers` + `CLOSE HELD`	Close throw is made right now. Centre does nothing. PRESS on that page pulses a throw through the same path as the hardware.
Green pip + `Motor current load — ACS712` + `0.00 A`	Live amps at rest.
Grey pip + `Motor PWM and direction` + `IDLE`	Duty zero. Grey pip is idle, not a fault.
Amber pip + `Jog by time` + `no encoder`	Amber because this page can command the motor. Timed detents. Shaft encoder is not read.
Amber pip + `Jog by position` + `needs enc` or red `locked`	Count detents. Green magnet required. Locked if I^2C or magnet is unhealthy.
Amber pip + `Obstruction test` + `MOVES MOTOR`	Same warning. Plastic-bottle test. Does not write the learned table.
`TURN scroll` · `PRESS open` · `HOLD exit`	Window the list, enter the highlighted item, or return to Main.

Full list:

#	Menu item	Value column	Opens
7	Jog by time	no encoder	Milliseconds per detent. Encoder unused.
8	Jog by position	needs enc / locked	Counts per detent. Requires a healthy AS5600.
9	Obstruction test	moves motor	Plastic-bottle envelope test.
10	Obstruction table	armed / stale	Close-direction graph first. TURN swaps to open.
11	EC11 knob and button	ok	Quadrature, detent scale, display timeout remaining.
12	Network and OTA	STA or AP	Address, mDNS, antenna note.
13	Stored calibration	ok / setup	Which steps are valid and why a stale one failed.

Every detail page keeps a pin strip: GPIO, terminal, function, live level or raw value. HOLD back returns to this menu. Bottle key, limit markers, and override rockers can inject a cooked signal from PRESS. The pin strip stays the real GPIO.

2. Bottle door key — 49E

Present and absent are two calibrated levels with separate enter and exit thresholds. Polarity is stated in words because it is configurable. PRESS can pulse the cooked state without moving the bottle.

Figure 2. DIAG · bottle key. Present, last edge 3 min 11 s ago.

On screen	Meaning
`DIAG · bottle key`	Header left.
`1 / 14`	Header right. First of fourteen detail screens. The menu is not numbered here.
`PRESENT` (green)	Hero. Cooked state after hysteresis. Amber plus `SIM` when PRESS has forced the state.
`pulse · adc 2971 · last edge …`	Selected action (pulse or recapture), live ADC, and age of the last cooked edge. Sparkline is the real ADC, newest at the right.
Sparkline + `adc · 8 s`	Raw ADC, newest at the right (dot). About 8 seconds once the strip is full. The trace scrolls left. The step up is the bottle going in.
`absent level (calibrated)` · `1 605`	Stage 1c absent capture.
`present level (calibrated)` · `2 980`	Stage 1c present capture.
`enter / exit threshold` · `2 400 / 2 100`	Hysteresis pair derived from those levels.
`trigger polarity` · `absent → OPEN`	Automatic Main command when the key leaves the recess.
Pin strip: `GPIO01` · `T23` · `ADC1_CH0` · `49E OUT`	MCU pin, Board A terminal, ADC channel, sensor output name. Always the real pin, even during a pulse.
`2 971 / 4095 · 2.39 V`	Live raw and volts at the pin.
`TURN action`	Pulse the key, or recapture levels. Pulse is the default. The highlight stops at each end. Extra clicks do not wrap.
`PRESS pulse`	Toggle cooked present/absent. Fires the same open/close edge as the hall sensor. Does not fake the ADC.
`PRESS recapture`	After TURN: capture absent, then PRESS again with the bottle in. Same two-step as Calibration 1c.
`HOLD back`	Return to the diagnostic menu. Clears a simulated key state.

3. Motor rotary encoder — AS5600

The only source of door position. It also carries the jitter band and the drift from the last marker re-sync. I²C health and magnet STATUS are separate: `ACK` with `no mag` on the menu means the chip answered and the magnet is unreadable.

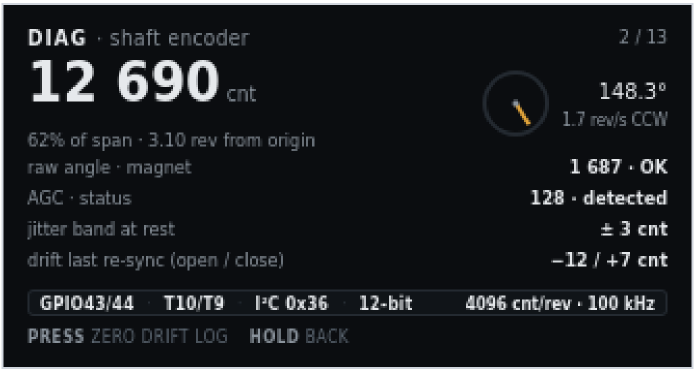

Figure 3. DIAG · shaft encoder. 62% of span, magnet OK.

On screen	Meaning
`DIAG · shaft encoder`	Header left.
`2 / 14`	Header right.
`12 690 cnt`	Hero. Accumulated counts from the open origin.
`62% of span · 3.10 rev from origin`	Share of measured span, and shaft revolutions since origin.
Angle dial	Single-turn AS5600 angle inside the current revolution. Not the multi-turn door position.
`148.3°`	That single-turn angle in degrees.
`1.7 rev/s CCW`	Live shaft speed and direction.
`I2C 0x36` · `ACK` or `NACK`	Live bus. ACK means the last STATUS/angle read succeeded. Magnet can still be unreadable.
`raw angle · magnet` · `1 687 · detected`	12-bit raw angle (0–4095) and STATUS: detected, lost, too weak, too strong, or unread if the bus failed.
`AGC (mid ~128)` · `128`	Automatic gain. Meaningful only on ACK. Mid-range is a healthy air gap.
`jitter band at rest` · `± 3 cnt`	Stage 1b capture. Growing band is an early warning for a weak magnet.
`drift last re-sync (open / close)` · `−12 / +7 cnt`	Correction applied the last time each marker was seen. Growing drift means lost turns or a slipping magnet.
Pin strip: `GPIO43/44` · `T10/T9` · `I²C 0x36` · `12-bit`	SCL/SDA, terminals, bus address, resolution.
`4096 cnt/rev · 100 kHz`	Counts per mechanical revolution and I²C clock.
`PRESS zero drift log`	Clear the displayed per-marker drift totals. Does not change origin.
`HOLD back`	Return to the menu.

4. Limit switches

HIGH is the marker state. A cut wire is indistinguishable from a made switch, so the screen says so. Both markers active is a fault, shown here rather than as a silent refusal to start. PRESS can force a marker without blocking the optical path.

Figure 4. DIAG · limit markers. Both clear, mid travel.

On screen	Meaning
`DIAG · limit markers`	Header left.
`3 / 14`	Header right.
`Open marker · GPIO17 · T7`	Left tile title. Pin and Board A terminal.
Grey pip + `CLEAR LOW`	Open switch not made. Electrical level is LOW.
`band 0 – 410 cnt · debounce 3`	Count window where an open-marker actuation is accepted, and debounce samples.
`Close marker · GPIO03 · T21`	Right tile title.
Grey pip + `CLEAR LOW`	Close switch not made.
`band 19 770 – 20 180 · deb 3`	Close-marker acceptance window and debounce.
`polarity · HIGH = marker or cut wire`	NC to GND with pull-up. HIGH means the loop is open: at the end, or the wire is cut.
`both active check · pass`	Both markers made at once is a fault. Pass means that is not happening.
`position now · 12 690 cnt · mid travel`	Live counts and a cooked region name.
`overtravel cap past marker · 220 cnt`	Hard cap after a marker. A seek that passes this is a fault.
Pin strip: `NC → GND · INPUT_PULLUP · kLimitActiveLow = false`	Wiring and the firmware polarity constant.
`17: LOW · 03: LOW`	Live GPIO17 and GPIO3 levels. Stays the real pin during a force.
`TURN select · PRESS force`	Highlight open or close. The highlight stops at each end. Extra clicks do not wrap. PRESS latches that marker as if the switch were made. Amber pip and `SIM` on the tile. PRESS again clears that force. A forced destination marker will stop a move already in progress.
`HOLD back`	Return to the menu. Clears a forced marker.

5. Override rockers

Two paralleled stations act as one logical switch. Each throw is shown separately. A stuck throw is invisible if you only display the combined result. Both throws at once is a defined tie-break, not a fault. PRESS pulses the selected throw through the same path as the hardware.

Figure 5. DIAG · override rockers. Close throw held.

On screen	Meaning
`DIAG · override rockers`	Header left.
`4 / 14`	Header right.
`Open throw · GPIO10 · T20`	Left tile. Open side of both paralleled stations.
Grey pip + `OFF HIGH`	Open throw not made. Pulled up.
`edges 14 · last 00:02:41`	Accepted edges since boot, and age of the last open edge.
`Close throw · GPIO18 · T8`	Right tile.
Green pip + `MADE LOW`	Close throw is held to ground right now.
`edges 11 · last 00:00:03`	Close-edge count and the edge three seconds ago.
`command model · edge trigger · centre does nothing`	Rocker centre is idle. Hold-to-run is not the model.
`both throws made · tie-break: OPEN wins`	If both throws hit in the same cycle, firmware commands open.
`debounce · 25 ms`	Switch debounce. Same family as Settings → Triggers.
`last accepted command · CLOSE · ignored, already closed`	Last edge that passed debounce, and why it did not start a move.
Pin strip: `active-low → GND · INPUT_PULLUP · 2 stations`	Wiring. Two stations in parallel hide a stuck mate unless this page splits the throws.
`10: HIGH · 18: LOW`	Live GPIO10 and GPIO18. Stays the real pin during a pulse.
`TURN select · PRESS pulse`	Highlight open or close. The highlight stops at each end. Extra clicks do not wrap. PRESS injects a short made-edge on that throw, the same command as the physical rocker. It can start a move. Amber pip and `SIM` while the pulse is live.
`HOLD back`	Return to the menu. Drops a pulse that has not expired.

6. Motor current load — ACS712-05B

Live amps against the learned envelope for the bin the door is in. Trip is a time-plus-threshold decision, so inrush skip and hold time are on the same page.

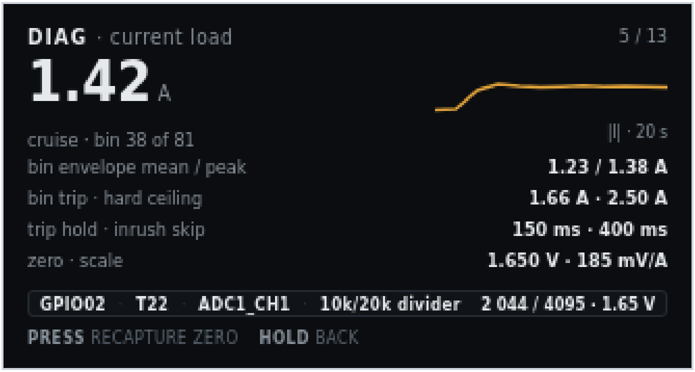

Figure 6. DIAG · current load. Live 1.42 A in cruise.

On screen	Meaning
`DIAG · current load`	Header left.
`5 / 14`	Header right.
`1.42 A`	Hero. Cooked motor current.
`cruise · bin 38 of 81`	Motion phase and which obstruction bin the door is in.
Sparkline + `|I| · 8 s`	Absolute current, newest at the right (dot). About 8 seconds once the strip is full. The trace scrolls left. The step up is the start of cruise.
`bin envelope mean / peak` · `1.23 / 1.38 A`	Learned mean and peak for bin 38.
`bin trip · hard ceiling` · `1.66 A · 2.50 A`	This bin's trip, then the compile-time ceiling.
`trip hold · inrush skip` · `150 ms · 400 ms`	How long current must stay over trip, and how long after PWM-on the envelope is ignored.
`zero · scale` · `1.650 V · 185 mV/A`	Stage 1a zero and the fixed ACS712-05B scale.
Pin strip: `GPIO02` · `T22` · `ADC1_CH1` · `10k/20k divider`	Pin, terminal, ADC channel, Board A divider.
`2 044 / 4095 · 1.65 V`	Live raw and volts. Near the rest zero, plus load.
`PRESS recapture zero`	Jump to Calibration stage 1a. Motor must be at rest.
`HOLD back`	Return to the menu.

7. Motor PWM and direction

Duty and K1 are separate blocks. Duty must reach zero and current must decay before the relay moves, so the blank timers are on this page.

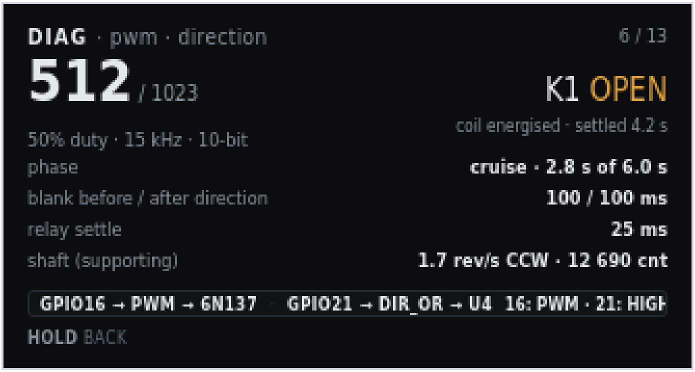

Figure 7. DIAG · pwm · direction. 50% duty, K1 open.

On screen	Meaning
`DIAG · pwm · direction`	Header left.
`6 / 14`	Header right.
`512 / 1023`	Hero. Live duty in 10-bit counts.
`50% duty · 15 kHz · 10-bit`	Percent, carrier, resolution. Carrier is compile-time. It is not a setting.
`K1 OPEN` (amber)	Direction relay commanded open. Amber because the coil is energised.
`coil energised · settled 4.2 s`	K1 has been in this state long enough to be settled.
`phase` · `cruise · 2.8 s of 6.0 s`	Profile phase and how much of cruise time has elapsed.
`blank before / after direction` · `100 / 100 ms`	PWM-off windows around a K1 change. Never reverse K1 without this blanking.
`relay settle` · `25 ms`	Wait after K1 before PWM may rise.
`shaft (supporting)` · `1.7 rev/s CCW · 12 690 cnt`	Encoder as a supporting signal on this page. The focused page is PWM.
Pin strip: `GPIO16 → PWM → 6N137` · `GPIO21 → DIR → U4`	PWM path onto Board B, and direction into U4.
`16: PWM · 21: HIGH`	GPIO16 is toggling. GPIO21 is HIGH for open on this build.
`HOLD back`	Return to the menu. This page does not command the motor.

8. Jog by time

Owns the motor. Each EC11 detent (or web − / +) queues a fixed run time at jog duty. Same-direction clicks add to that queue. Opposite clicks subtract; when the queue is gone, that click reverses through the normal PWM-off, decay, blank, and K1 settle windows. PRESS while idle cycles duty (5–25%) and time per detent (200–1000 ms). **While that timed move is running**, the shaft encoder is not polled, so I^2C NACKs cannot stretch the queue. Idle on this page still reads the AS5600 (same as most diagnostic screens). After PWM stops, one catch-up read refreshes STATUS. Markers and the learned envelope do not start, stop, or abort the move. Current is a live 0–2.50 A graph when the ACS712 is sampling; it is display-only. The pin strip reads `encoder ignored`.

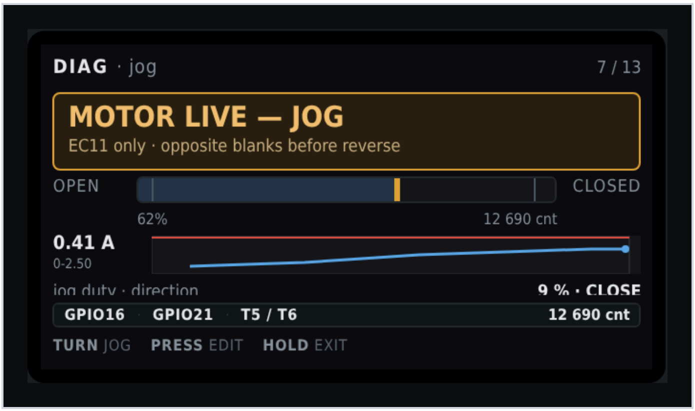

Figure 8. DIAG · jog by time. Timed queue. Live firmware titles the page MOTOR LIVE — TIME and omits the travel widget.

On screen	Meaning
`DIAG · jog by time`	Header left.
`7 / 14`	Header right.
`MOTOR LIVE — TIME`	Amber banner. This page owns the motor. Hall and rocker cannot preempt it.
`encoder unused — time per detent`	Banner detail. AS5600 I²C is skipped only while the timed queue is moving, not while this page is idle.
`jog duty · direction` · `9 % · CLOSE`	Jog PWM as a percent of 10-bit full scale (5–25%). K1 for this jog. PRESS while idle highlights this row; TURN changes it.
`step · remaining` · `400 ms · 800 ms`	Time added per detent (200–1000 ms, 50 ms steps) and how much powered time is still queued. Queue caps at 4 s. Remaining is wall-clock, not loop ticks.
Live amp strip · 0–2.50 scale	ACS712 history. Display-only. Not used to run or stop the jog.
Pin strip: `GPIO16` · `GPIO21` · `encoder ignored`	PWM and direction only. No shaft counts on this page.
`TURN jog` or `TURN duty` / `TURN step`	Default: CW queues close, CCW queues open. Same way extends. Opposite shortens, then blanks and reverses.
`PRESS stop` or `PRESS edit` / `PRESS next`	While moving: drop PWM on a short click (after debounce, on release). While idle: cycle Jog → duty % → step ms → Jog.
`HOLD exit`	Stop if moving, save a pending edit, then return to the menu.

9. Jog by position

Owns the motor only while the AS5600 ACKs and STATUS shows a magnet that is neither too weak nor too strong. That is not a trusted origin; calibration is not required. The queue is **relative**: each detent is shaft counts in the commanded direction from the live unwrap, not distance from open or close. The pin strip may still show accumulated counts from origin when origin is trusted. Opening this page cancels a leftover time-jog move and forces an encoder poll so the lock banner is not a stale STATUS from a skipped I²C window. Each detent uses the same jog duty (default 64 counts, 16–512 in steps of 16). Same-direction clicks extend the count queue (cap 2048). Opposite clicks subtract, then reverse through the same blanking path as time jog. The stall watchdog is armed when PWM starts so an old timer cannot trip on the first frame. Travel-cap (absolute) is not applied on this page. If I²C fails or STATUS drops out of range, PWM stops immediately without latching a sensor fault. While locked, the banner title is `ENCODER REQUIRED` and the subtitle names the check that failed. No dedicated LCD shot in this set; chrome matches time jog with counts in place of milliseconds.

On screen	Meaning
`DIAG · jog by position`	Header left.
`8 / 14`	Header right.
`MOTOR LIVE — COUNTS`	Amber banner when I²C ACK and STATUS are good. TURN jogs.
`stops if encoder is lost`	Banner detail when the sensor is good.
`ENCODER REQUIRED`	Amber banner. Jog is locked. TURN does not move the motor.
`I2C NACK - chip unread at 0x36`	Banner detail. The AS5600 did not ACK. Not a magnet STATUS flag. Bus time is the 10 ms Wire timeout on a failed transfer; too-weak/too-strong do not change that time.
`STATUS: magnet too weak`	Banner detail. Chip ACKed. STATUS bit "too weak." Angle was read and discarded.
`STATUS: magnet too strong`	Banner detail. Chip ACKed. STATUS bit "too strong."
`STATUS: magnet lost`	Banner detail. Chip ACKed. Magnet-detect bit clear (and not the too-weak/too-strong pair).
Travel widget	Shown. Live counts and magnet label on the pin strip (`AS5600 T5 / T6`).
`jog duty · direction`	Same 5–25% duty as time jog. Shared setting.
`step · remaining` · counts	Relative counts per detent and remaining in the commanded direction. Not a target absolute position.
`TURN jog`	Ignored while locked. Otherwise same CW close / CCW open as time jog.
`PRESS stop` / `HOLD exit`	Same cancel rules as time jog.

10. Obstruction test

Plastic bottle between the doors. Same envelope rules as Main. Does not write the learned table. Learn runs assume a clear path. This test is not a substitute for them.

Figure 9. DIAG · obstruction test. Close trip, reversed, opening.

On screen	Meaning
`DIAG · obstruction test`	Header left.
`9 / 14`	Header right.
`TRIPPED — REVERSED`	Amber banner title. Close obstruction fired. Reverse is in progress or just finished.
`Close obstruction at bin 41 · 2.11 A vs trip 1.65 A`	Where and how far over trip.
`test load` · `plastic bottle between doors`	Specified test article. Not a body part, not a tool.
`expected close / open` · `reverse once / fault first trip`	Close reverses once. Open faults on the first trip. Same as Main.
`result this run` · `reversed, then opened to marker`	What this test actually did.
`table writes · hard ceiling` · `disabled · 2.50 A armed`	A deliberate obstruction cannot be taught as load. The ceiling stays on.
Pin strip: `GPIO02 current` · `GPIO16 pwm` · `GPIO21 dir`	Signals this test uses.
`1.02 A · opening`	Live current and direction after reverse.
`PRESS run again`	Start another close into the bottle.
`HOLD exit`	Cancel if live, then return to the menu.

11. Obstruction table — close direction

Learned envelope as a curve. Breakaway at the open end and seal loading at the close end are the shape you want. A bump in the middle is a bind. The grey block past the close marker is travel the learn runs never enter, because snug is a separate window.

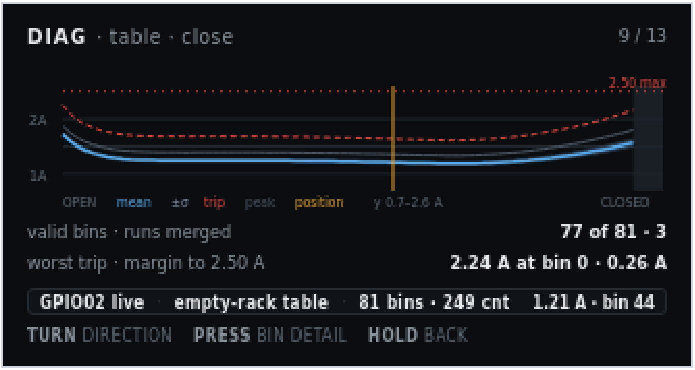

Figure 10. DIAG · table · close. Empty-rack table, live at bin 44.

On screen	Meaning
`DIAG · table · close`	Header left. Close-direction envelope.
`10 / 14`	Header right. Menu item 10 is this pair of graphs.
`1A` / `2A` on the Y axis	Current scale. The window is 0.7–2.6 A so the curve is readable on 170 pixels.
Dashed red line + `2.50 max`	Compile-time hard ceiling. Not a learned value.
Blue `mean` curve	Merged mean current per bin from the learn runs.
Grey band `±σ`	One-sigma band around the mean.
Dashed red `trip` curve	Per-bin trip used on Main.
Thin grey `peak` curve	Highest sample merged into that bin.
Vertical amber `position` line	Live door position on this graph.
Dark block near `CLOSED`	Bins past the close marker. Learn runs never visit them. Snug is a separate window.
Legend: `OPEN` · `mean` · `±σ` · `trip` · `peak` · `position` · `y 0.7–2.6 A` · `CLOSED`	X ends and the plotted series. Y window is stated so the clipped axis is not mistaken for 0–2.50 A.
`valid bins · runs merged` · `77 of 81 · 3`	Bins with data, and how many strokes were merged.
`worst trip · margin to 2.50 A` · `2.24 A at bin 0 · 0.26 A`	Highest trip on this table (breakaway at the open end) and room left to the ceiling.
Pin strip: `GPIO02 live` · `empty-rack table` · `81 bins · 249 cnt`	Which table is drawn, bin count, and bin width in counts.
`1.21 A · bin 44`	Live current and the bin under the position line.
`TURN direction`	Swap to the open-direction graph.
`PRESS bin detail`	Numeric values for the bin under the cursor (not a separate mockup in this set).
`HOLD back`	Return to the menu.

12. Obstruction table — open direction

Same table, other direction. Breakaway is now at the closed end, where the door pulls out of the seal. The unvisited block moves to the open end. Two independent tables are why both views exist.

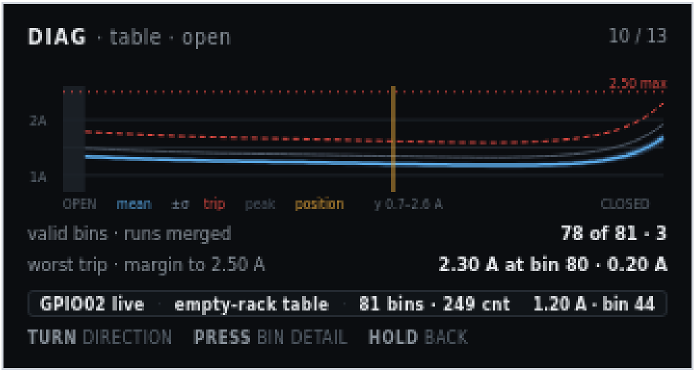

Figure 11. DIAG · table · open. Breakaway at the closed end.

On screen	Meaning
`DIAG · table · open`	Header left.
`11 / 14`	Header right.
Graph, Y labels, `2.50 max`, legend	Same series as the close graph. Shape differs: peak at the closed end.
Dark block near `OPEN`	Bins before the open marker. Open-direction learn runs never visit them.
`valid bins · runs merged` · `78 of 81 · 3`	Open-direction bins with data (bins at and after the open marker) and merged run count.
`worst trip · margin to 2.50 A` · `2.30 A at bin 80 · 0.20 A`	Highest open-direction trip (breakaway at the closed end) and room left to the ceiling.
Pin strip: `GPIO02 live` · `empty-rack table` · `81 bins · 249 cnt`	Same table family as the close view.
`1.20 A · bin 44`	Live current and the bin under the position line, read against the open envelope.
`TURN direction` · `PRESS bin detail` · `HOLD back`	Same gestures as the close graph.

13. EC11 knob and button

An input with no other home. Detent scale is bench-calibrated, so raw quadrature and the scaled count sit side by side.

Figure 12. DIAG · ec11. Detent scale 100 / 400. Display timeout 5 min.

On screen	Meaning
`DIAG · ec11`	Header left.
`12 / 14`	Header right.
`+37`	Hero. Scaled detents since reset. Sign is CW positive on this shot.
`detents since reset · CW`	What the hero counts, and last turn sense.
`SW released`	Knob switch is up.
`last press 180 ms`	Duration of the last PRESS (under the HOLD threshold).
`raw quadrature` · `134 counts`	Unscaled CLK/DT edges. Compare with the detent hero to check the scale.
`detent scale` · `100 / 400`	Quadrature-to-detent fraction from `ec11_calibration.h` (`kEc11DetentScaleNumerator` / `kEc11DetentScaleDenominator`). 100/400 is one detent per four edges.
`debounce` · `25 ms`	SW debounce.
`display timeout` · `5 min · 3:41 remaining`	Idle blanking setting and time left until GPIO38 backlight and GPIO15 panel power drop. Default is 5 minutes.
Pin strip: `GPIO11 CLK` · `GPIO12 DT` · `GPIO13 SW` · `on-board`	EC11 on the T-Display-S3, not a Board A terminal.
`1 · 0 · HIGH`	Live CLK, DT, SW. SW HIGH is released with pull-up.
`PRESS zero count`	Reset the detent hero to 0. Does not change calibration constants.
`HOLD back`	Return to the menu.

14. Network and OTA

Setup surface. The steel lid can block the onboard antenna, so that note is on the screen, not only in the build docs. When SoftAP is serving, this page shows the join password in clear (default `reserved`). The AP name defaults to `PrivateReserve`. After join, open `http://192.168.4.1/`. PRESS rejoin retries the stored STA SSID. To drop that SSID, use Settings → Service → Clear station Wi-Fi, then reboot.

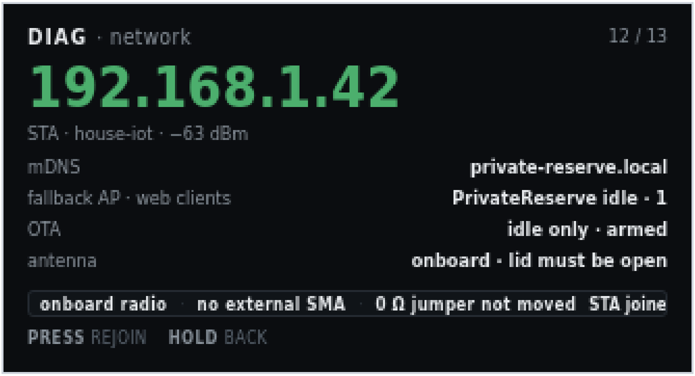

Figure 13. DIAG · network. STA joined, OTA idle-only.

On screen	Meaning
`DIAG · network`	Header left.
`13 / 14`	Header right.
`192.168.1.42` (green)	Hero. Station address when STA joined.
`STA · house-iot · −63 dBm`	Role, stored SSID, RSSI.
`mDNS` · `private-reserve.local`	mDNS name for HTTP on the joined LAN.
`fallback AP · web clients` · `PrivateReserve idle · 1`	SoftAP name, idle because STA joined, and one HTTP client. If SoftAP is the live link, this row is `join password` and the WPA2 PSK (default `reserved`) instead.
`OTA` · `idle only · armed`	OTA is allowed only while the motor is idle. Armed means the endpoint is listening.
`antenna` · `onboard · lid must be open`	No external SMA. Close the steel lid and RSSI can collapse.
Pin strip: `onboard radio` · `no external SMA` · `0 Ω jumper not moved`	Antenna path as shipped. Do not move the 0 Ω jumper unless you fit an external antenna.
`STA joined`	Live radio role.
`PRESS rejoin`	Retry STA join with the stored SSID. If join fails, SoftAP starts. This does not clear the stored password. To wipe station credentials, use Settings → Service → Clear station Wi-Fi, then reboot from that result screen or from Reboot controller.
`HOLD back`	Return to the menu.

15. Stored calibration

Answers whether this controller may run automatically, and why not. Stale entries name the value that invalidated them.

Figure 14. DIAG · calibration. Loaded table stale after cruise duty 512 → 544.

On screen	Meaning
`DIAG · calibration`	Header left.
`14 / 14`	Header right. Last detail screen.
Green pip + `Direction · origin · span` + `OK`	Stages 2–4 stored and valid.
Green pip + `Marker edges (both)` + `± 6 CNT`	Open/close probe spread still inside tolerance. The value column shows the open-side figure on this shot.
Green pip + `Closed seated · by current` + `OK`	Stage 5 method still current-rise.
Green pip + `Stroke 32.0 in → 631 cnt/in` + `OK`	Stage 6 still matches the stored span.
Green pip + `Obstruction table — empty racks` + `ARMED`	Empty envelope is in service.
Amber pip + highlight + `Obstruction table — loaded racks` + `STALE`	Selected. Loaded table is not usable.
Grey pip + `stale since cruise duty 512 → 544`	Why. A motion-profile change invalidates both tables. This row names the field that did it.
Pin strip: `NVS` · `format v3` · `written 41 boots ago`	Where the blob lives, format, age in boots (no RTC).
`3.6 kB blob`	Stored size.
`PRESS open`	Jump to the matching Calibration stage for the highlighted row.
`HOLD back`	Return to the menu.

16. Related

- Software overview — full web UI, figure 1
- Settings — clear station Wi-Fi and reboot
- Board A — T7, T8, T9/T10, T20–T23

This page is a chapter of the Software User Manual.